

- 13** A pair of twins of equal masses travels on separate express trains that run along the Earth's equator. Edison is on the train travelling eastwards while Wilson is on the train travelling westwards.

Assuming both trains are moving at equal speed (relative to the Earth's surface), which of the following statements about their measured weight is correct?

- A** Edison weighs more as his motion is in the same direction to the Earth's rotational motion about its axis.
- B** Wilson weighs more as his motion is counter to the Earth's rotational motion about its axis.
- C** Both weigh more than when the train is stationary since kinetic energy is converted to an increase in mass.
- D** Both weigh the same since motion along the Earth's equator does not affect their measured weights.

13	B	Earth rotates to the east, so Edison will have higher angular velocity while Wilson will have lower angular velocity. $mg - N = mr\omega^2$ N will give the balance reading. $N = mg - mr\omega^2$ So Wilson will have higher reading.
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14 The circuit shown contains a cell of negligible internal resistance, a variable resistor of